(i)

For L_a,

$$y - 0 = \frac{(1-a)-0}{0-a}(x-a)$$

$$y = -\frac{1-a}{a}x + 1 - a$$

For L_b,

$$y = -\frac{1-b}{b}x + 1 - b$$

$$y = -\frac{1-a}{a}x + 1 - a = -\frac{1-b}{b}x + 1 - b$$

$$\frac{b-a}{ab}x = b - a$$

$\because a \neq b, \quad \therefore x = ab$

$$y = (1-a)(1-b)$$

Points of intersection: $(ab,\ (1-a)(1-b))$

(ii)

$$b \to a \implies (ab,\ (1-a)(1-b)) \to (a^2,\ (1-a)^2)$$

Let $x = a^2 \quad (0 < a < 1)$,

$$y = (1-a)^2 = (1-\sqrt{x})^2$$

(iii)

$$C:\ y = (1-\sqrt{x})^2 = 1 - 2\sqrt{x} + x$$

$$\frac{dy}{dx} = 1 - \frac{1}{\sqrt{x}} = 1 - \frac{1}{a} = -\frac{1-a}{a}$$

$$y - (1-a)^2 = -\frac{1-a}{a}(x - a^2)$$

$$y = -\frac{1-a}{a}x + 1 - a$$

$\therefore L_a$ is tangent to C at $(a^2, (1-a)^2)$

This solution was submitted by **Jon**

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